

1.

```
CREATE DATABASE CollegeDB;  
USE CollegeDB;
```

```
CREATE TABLE Student (  
    Student_ID INT PRIMARY KEY,  
    Student_Name VARCHAR(50),  
    Course VARCHAR(50)  
);
```

```
CREATE TABLE Student_Count (  
    Total_Students INT  
);
```

```
INSERT INTO Student_Count VALUES (0);
```

```
DELIMITER //
```

```
CREATE TRIGGER after_student_insert  
AFTER INSERT ON Student  
FOR EACH ROW  
BEGIN  
    UPDATE Student_Count  
    SET Total_Students = Total_Students + 1;  
END //
```

```
DELIMITER ;
```

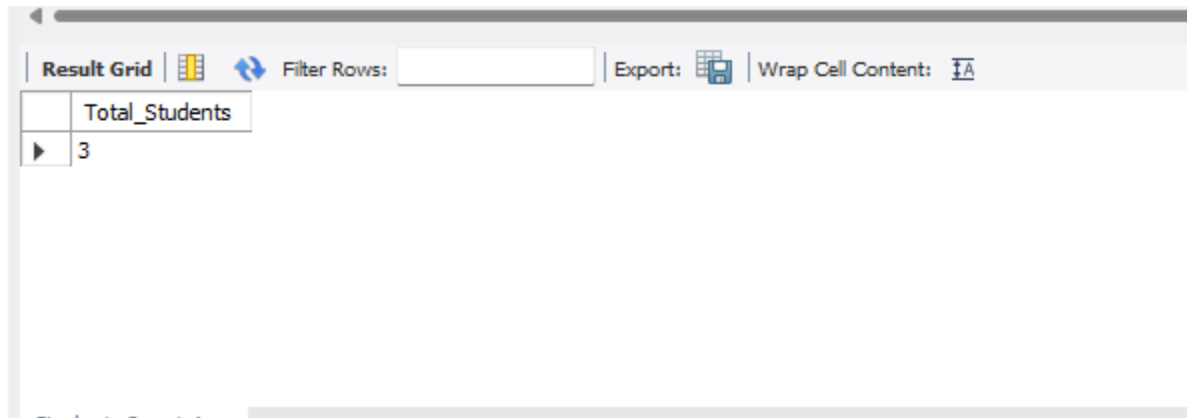
```
INSERT INTO Student VALUES  
(101, 'Spooorthi', 'BCA');
```

```
INSERT INTO Student VALUES  
(102, 'Neha', 'BCA');
```

```
INSERT INTO Student VALUES  
(103, 'Manya', 'BCA');
```

```
SELECT * FROM Student;
```

```
SELECT * FROM Student_Count;
```



The screenshot shows a database query result grid. The grid has a single column labeled 'Total_Students' and a single row with the value '3'. The interface includes a toolbar with options like 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'.

Total_Students
3

2.

```
CREATE DATABASE CompanyDB;  
USE CompanyDB;
```

```
CREATE TABLE Employee (  
    Emp_ID INT PRIMARY KEY,  
    Emp_Name VARCHAR(100),  
    Department VARCHAR(100),  
    Salary DECIMAL(10,2)  
);
```

```
CREATE TABLE Salary_Audit (  
    Audit_ID INT AUTO_INCREMENT PRIMARY KEY,  
    Emp_ID INT,  
    Old_Salary DECIMAL(10,2),  
    New_Salary DECIMAL(10,2),  
    Change_Time TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

```
INSERT INTO Employee VALUES  
(1, 'Spoorthi', 'IT', 50000);
```

```
INSERT INTO Employee VALUES  
(2, 'Neha', 'HR', 45000);
```

```
DELIMITER //
```

```
CREATE TRIGGER after_salary_update
```

```

AFTER UPDATE ON Employee
FOR EACH ROW
BEGIN
    IF OLD.Salary <> NEW.Salary THEN

        INSERT INTO Salary_Audit
        (Emp_ID, Old_Salary, New_Salary)
        VALUES
        (OLD.Emp_ID, OLD.Salary, NEW.Salary);

    END IF;
END //

DELIMITER ;

UPDATE Employee
SET Salary = 60000
WHERE Emp_ID = 1;

UPDATE Employee
SET Salary = 50000
WHERE Emp_ID = 2;

SELECT * FROM Employee;

SELECT * FROM Salary_Audit;

```

Result Grid					
Filter Rows:		Edit:		Export/Import:	
				Wrap Cell Content:	
	Audit_ID	Emp_ID	Old_Salary	New_Salary	Change_Time
▶	1	1	50000.00	60000.00	2026-09-29 11:47:49
	2	2	45000.00	50000.00	2026-09-29 11:47:49
*	NULL	NULL	NULL	NULL	NULL

3.

```

DROP DATABASE IF EXISTS CompanyDB2;

CREATE DATABASE CompanyDB2;
USE CompanyDB2;

CREATE TABLE Employee (

```

```
Emp_ID INT PRIMARY KEY,  
Emp_Name VARCHAR(50),  
Department VARCHAR(50),  
Salary DECIMAL(10,2)  
);
```

```
CREATE TABLE Deleted_Employee_Log (  
Log_ID INT AUTO_INCREMENT PRIMARY KEY,  
Emp_ID INT,  
Emp_Name VARCHAR(50),  
Deleted_Time TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

```
INSERT INTO Employee VALUES (101, 'Spoorthi', 'IT', 50000);  
INSERT INTO Employee VALUES (102, 'Neha', 'HR', 45000);  
INSERT INTO Employee VALUES (103, 'Manya', 'Management', 80000);  
INSERT INTO Employee VALUES (104, 'Navya', 'Finance', 60000);
```

```
DELIMITER //  
CREATE TRIGGER before_employee_delete  
BEFORE DELETE ON Employee  
FOR EACH ROW  
BEGIN  
IF OLD.Department = 'Management' THEN  
SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Management employees  
cannot be deleted';  
END IF;  
END //  
DELIMITER ;
```

```
DELIMITER //  
CREATE TRIGGER after_employee_delete  
AFTER DELETE ON Employee  
FOR EACH ROW  
BEGIN  
INSERT INTO Deleted_Employee_Log (Emp_ID, Emp_Name) VALUES  
(OLD.Emp_ID, OLD.Emp_Name);  
END //  
DELIMITER ;
```

```
DELETE FROM Employee WHERE Emp_ID = 101;
DELETE FROM Employee WHERE Emp_ID = 104;
```

```
SELECT * FROM Employee;
SELECT * FROM Deleted_Employee_Log;
```

Result Grid				
Log_ID	Emp_ID	Emp_Name	Deleted_Time	
1	101	Spoorthi	2026-09-29 12:06:20	
2	104	Navya	2026-09-29 12:06:20	
NULL	NULL	NULL	NULL	

Employee 4 Deleted Employee Log5 x				
Output				
Action Output				
#	Time	Action	Message	
83	12:06:20	INSERT INTO Employee VALUES (104, 'Navya', 'Finance', 60000)	1 row(s) affected	
84	12:06:20	CREATE TRIGGER before_employee_delete BEFORE DELETE ON Employee FOR EACH ROW BEGIN IF OL...	0 row(s) affected	
85	12:06:20	CREATE TRIGGER after_employee_delete AFTER DELETE ON Employee FOR EACH ROW BEGIN INSERT I...	0 row(s) affected	
86	12:06:20	DELETE FROM Employee WHERE Emp_ID = 101	1 row(s) affected	
87	12:06:20	DELETE FROM Employee WHERE Emp_ID = 104	1 row(s) affected	
88	12:06:20	SELECT * FROM Employee LIMIT 0, 2000	2 row(s) returned	
89	12:06:20	SELECT * FROM Deleted_Employee_Log LIMIT 0, 2000	2 row(s) returned	

If we delete emp_id 103 it won't because manager cann't remove

```

44
45 • DELETE FROM Employee WHERE Emp_ID = 101;
46 • DELETE FROM Employee WHERE Emp_ID = 103;
47
48 • SELECT * FROM Employee;

```

Result Grid				
Log_ID	Emp_ID	Emp_Name	Deleted_Time	
1	101	Spoorthi	2026-09-29 12:11:13	
NULL	NULL	NULL	NULL	